Econometrics Chapter 4: Introduction to Matrix Algebra

(1) Matrix Definitions and Operations

Matrix

- matrix is a rectangular array of numbers, symbols, or expressions arranged in rows and columns

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$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{m1} & \cdots & \cdots & a_{mn} \end{bmatrix}$$

- order of dimensions: $m \times n$
- column vectors: $m \times 1$ —— assume x to be a colum vector for this
- row vectors: $1 \times n$

- a_{ij} is the (i,j) element or term of the matrix

$(A)_{ij}$ is the (i,j) element of the matrix A

$(a_{ij})_{m \times n}$ refers to a $m \times n$ matrix with (i,j)th element a_{ij}

Matrix Operations

1. Equality $A=B \Leftrightarrow (A)_{ij} = (B)_{ij}$ for all $i=1,\ldots,m$, $j=1,\ldots,n$
2. Addition: $(A+B)_{ij} = (A)_{ij} + (B)_{ij}$ for all $i=1,\ldots,m$, $j=1,\ldots,n$
3. Hadamard Product: $(A \odot B)_{ij} = (A)_{ij}(B)_{ij}$ [Element by Element Multiplication]
4. Scalar Multiplication: $(\alpha A)_{ij} = \alpha (A)_{ij}$ for all $i=1,\ldots,m$, $j=1,\ldots,n$

Properties:

a. $(A+B)+C = A+(B+C)$ and $(A \odot B) \odot C = A \odot (B \odot C)$

b. $A+B = B+A$ and $A \odot B = B \odot A$

c. $A \odot (B+C) = A \odot B + A \odot C$

d. $\alpha(A+B) = \alpha A + \alpha B$

$(\alpha+\beta)A = \alpha A + \beta A$

5. Matrix Transposition

- $A_{ij}^T = A_{ji}$

  can be written as A'

- symmetric matrix $A^T = A$

- Properties:

  a. $(A+B)^T = A^T + B^T$

  b. $(A \odot B)^T = A^T \odot B^T$

  c. $(\alpha A)^T = \alpha (A^T)$

  d. $(A^T)^T = A$

6. Matrix Multiplication

- $A_{m \times n}$ $B_{m \times p}$ [no. of col (A) = no. of row (B)]

  $(AB)_{ij} = \Sigma a_{ik} b_{kj}$

- Properties:

  a. $(AB)C = A(BC)$

  b. $A(B+C) = AB + AC$

  c. $(A+B)C = AC + BC$

  d. $AB = 0$ does not imply $A=0$ or $B=0$

  e. $Ab = Ac$ does not imply $b=c$

  f. $(AB)^T = B^T A^T$

7. Identity Matrix

$$(I_n)_{ij} \begin{cases} 1 \text{ if } i=j \\ 0 \text{ if } i \neq j \end{cases} \quad \text{for } i,j = 1, \ldots, n$$

8. Diagonal, Upper and Lower Triangle Matrix

- Diagonal: $\begin{bmatrix} * & \cdots & 0 \\ 0 & * & \vdots \\ \vdots & \ddots & \vdots \\ 0 & \cdots & * \end{bmatrix}$

- Lower Triangle: $\begin{bmatrix} * & 0 & \cdots & 0 \\ * & * & & \vdots \\ \vdots & & \ddots & \vdots \\ * & \cdots & & * \end{bmatrix}$

- Upper Triangle: $\begin{bmatrix} * & * & \cdots & * \\ 0 & * & \ddots & \vdots \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & \cdots & * \end{bmatrix}$

9. General Linear System of Equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots \qquad\qquad\qquad\qquad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

- can be expressed as $Ax=b$

whereby, $\underset{m\times n}{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}} \underset{n\times 1}{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}} = \underset{m\times 1}{\begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}}$

10. Inner Product : x^Tx (Dot Product)

$$x \cdot x = x^Tx = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$= \sum_{i=1}^{n} x_i^2$$

norm of a vector $= \sqrt{x^Tx}$ (single number that measures the length or magnitude of the vector)

- $x^Ty = 0 \iff$ x and y are orthogonal (geometrically: perpendicular)

11. Outer Product : xx^T

$$xx^T = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}$$

$$= \begin{bmatrix} x_1^2 & x_1x_2 & \cdots & x_1x_n \\ x_2x_1 & \ddots & & \vdots \\ \vdots & & \ddots & \\ x_nx_1 & \cdots & & x_n^2 \end{bmatrix}$$

12. Quadratic Form

$$x^T A x = [x_1 \; x_2 \; \dots \; x_n] \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{n1} & \cdots & & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= a_{11}x_1^2 + a_{22}x_2^2 + \dots + a_{nn}x_n^2 + x_1x_2(a_{12} + a_{21}) + \dots$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} x_i x_j a_{ij}$$

13. Trace of a matrix

- addition of all

$$tr(A) = \sum a_{ii}$$

$$tr(AB) = tr(BA)$$

$$tr(xx^T) = \sum x_i^2$$

$$= tr(x^T x)$$

(2) Inverse Matrix

Left inverse: B is a left inverse of A

Given $BA = I_n$

Right inverse: C is a right inverse of

Given $AC = I_n$

- A has both right and left inverse only if it is a square
  - it will be the same matrix
  - $BA = AC = I_n$, $B = C$
- $AA^{-1} = I_n = A^{-1}A$

System of Linear Equations

- if A has an inverse

unique sol^{n} to system: $Ax = b$

$x = A^{-1}b$

Formula to find inverse: $\frac{1}{|A|}\begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$ ——— related expansion

- $|A| = \det(A) = a_{11}a_{22} - a_{12}a_{21}$
- If inverse exist $|A| \neq 0$: A is non-singular
  if $|A| = 0$
  - one or more row or column of 0
  - one row or column is a multiple of the other (Linear Combinations)

- Gaussian Elimination
- Cofactor formula for A^{-1}
- Laplace Expansion for $\det(A)$

- Useful Results
  a. $(A^{-1})^T = (A^T)^{-1}$
  b. $(AB)^{-1} = B^{-1}A^{-1}$

(3) Partitioned Matrices [Block Matrices]

Partitioned Matrices

- mxn matrix that is partition into blocks of sub matrixes
- partitioned matrices are called block matrices

1. Addition of Partitioned Matrices

$$A = \begin{bmatrix} A_{11}_{m_1 \times n_1} & A_{12}_{m_1 \times n_2} \\ A_{21}_{m_2 \times n_1} & A_{22}_{m_2 \times n_2} \end{bmatrix} \quad B = \begin{bmatrix} B_{11}_{m_1 \times n_1} & B_{12}_{m_1 \times n_2} \\ B_{21}_{m_2 \times n_1} & B_{22}_{m_2 \times n_2} \end{bmatrix}$$

$$A+B = \begin{bmatrix} A_{11}+B_{11}_{m_1 \times n_1} & A_{12}+B_{12}_{m_1 \times n_2} \\ A_{21}+B_{21}_{m_2 \times n_1} & A_{22}+B_{22}_{m_2 \times n_2} \end{bmatrix}$$

2. Multiplication of Partitioned Matrices

$$AB = \begin{bmatrix} A_{11}_{m_1 \times n_1} & A_{12}_{m_1 \times n_2} \\ A_{21}_{m_2 \times n_1} & A_{22}_{m_2 \times n_2} \end{bmatrix} \begin{bmatrix} B_{11}_{n_1 \times p_1} & B_{12}_{n_1 \times p_2} \\ B_{21}_{n_2 \times p_1} & B_{22}_{n_2 \times p_2} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21}_{m_1 \times p_1} & A_{11}B_{12} + A_{12}B_{22}_{m_1 \times p_2} \\ A_{21}B_{11} + A_{22}B_{21}_{m_2 \times p_1} & A_{21}B_{12} + A_{22}B_{22}_{m_2 \times p_2} \end{bmatrix}$$

3. Transposition of Partitioned Matrix

$$A^T = \begin{bmatrix} A_{11}{}^T & A_{12}{}^T \\ A_{21}{}^T & A_{22}{}^T \end{bmatrix}$$

4. Partitioned into Columns

$$X_{*i} = [x_{*1} \quad x_{*2} \quad \dots \quad x_{*k}]$$

with each x_{*i} being a column vector

5. Partitioned into Rows

$$X_{i*} = \begin{bmatrix} x_{1*} \\ x_{2*} \\ \vdots \\ x_{n*} \end{bmatrix}$$ with each x_{i*} being a row vector

Properties: $X^T X = \sum X_{i*}{}^T X_{i*}$

6. Inverse of a Partition Matrix

$$A = \begin{bmatrix} A_{11}_{m_1 \times n_1} & A_{12}_{m_1 \times n_2} \\ A_{21}_{m_2 \times n_1} & A_{22}_{m_2 \times n_2} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} & -(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1}A_{12} \\ -A_{22}^{-1}A_{21}(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} & A_{22}^{-1} + A_{22}^{-1}A_{21}(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1}A_{12}A_{22}^{-1} \end{bmatrix}$$

$$A^{-1}A = \begin{bmatrix} I_{m_1} & 0_{m_1 \times m_2} \\ 0_{m_2 \times m_1} & I_{m_2} \end{bmatrix}$$

7. Kronecker Product ⊗

- mxn A with a pxq B

Results: mpxnq block matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \left[\begin{array}{cc|cc|cc} a_{11} & 0 & a_{12} & 0 & a_{13} & 0 \\ 0 & a_{11} & 0 & a_{12} & 0 & a_{13} \\ \hline a_{21} & 0 & a_{22} & 0 & a_{23} & 0 \\ 0 & a_{21} & 0 & a_{22} & 0 & a_{23} \end{array}\right]$$

(4) Vectors and Matrices of Random Variables

1. x

- allow x to be an $m \times 1$ vector of random variables

$$x = [X_1 \; X_2 \; \dots \; X_m]^T$$

- $E(x) = [E(X_1) \; E(X_2) \; \dots \; E(X_m)]^T$

2. X_i

- allow X to be an $m \times n$ matrix of random variables

reading set 1 / reading set 2

$$X_i = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1n} \\ X_{21} & X_{22} & \dots & X_{2n} \\ \vdots & \vdots & & \vdots \\ X_{m1} & X_{m2} & \dots & X_{mn} \end{bmatrix}$$

$$E(X_i) = \begin{bmatrix} E(X_{11}) & \dots & E(X_{1n}) \\ \vdots & & \\ E(X_{m1}) & \dots & E(X_{mn}) \end{bmatrix}$$

partitioned by columns

$$X_i = [X_{*1} \; X_{*2} \; \dots \; X_{*n}]$$

whereby,

$$E(X_i) = [E(X_{*1}) \; E(X_{*2}) \dots \; E(X_{*n})]$$

such that $E(X_{11}) = E(X_{m1})$

Explanation:

Let small $x = (X_1, ..., X_m)'$ be an $m \times 1$ vector of random variables representing the independent variables (e.g., Wind, Rainfall, etc.).
Then

$$E(x) = (E[X_1], ..., E[X_m])'$$

is the vector of their expectations.

Now, consider each big X_i (e.g., Wind) as an $m \times n$ **matrix of random variables** representing multiple readings of the same variable across n different instances (e.g., days, firms, or time periods):

$$X_i = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1n} \\ X_{21} & X_{22} & \dots & X_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ X_{m1} & X_{m2} & \dots & X_{mn} \end{bmatrix}.$$

If we partition X by **columns**,

$$X = [X_{*1}, X_{*2}, ..., X_{*n}],$$

then each column X_{*j} represents one group of readings of the same variable across all observations.

Since these readings are assumed to come from the same distribution,

$$E(X_{11}) = E(X_{21}) = \cdots = E(X_{m1}) = E(X_{*1}).$$

In words: within each column (e.g., wind readings over time), all elements share the same expectation because they are identically distributed realizations of the same random variable.

3. Mean Centering of x

- let x be a $m \times 1$ vector of random variables

$$\tilde{x} = x - E(x)$$

$$= \begin{bmatrix} X_1 - E(X_1) \\ X_2 - E(X_2) \\ \vdots \\ X_m - E(X_m) \end{bmatrix} = \begin{bmatrix} \tilde{X}_1 \\ \tilde{X}_2 \\ \vdots \\ \tilde{X}_m \end{bmatrix}$$

4. Variance Covariance Matrix

- $\text{Var}(x) = E(x - E(x))E((x - E(x))^T$

$$= E(\tilde{x}\tilde{x}^T)$$

$$= E\begin{bmatrix} \tilde{X}_1^2 & \tilde{X}_1\tilde{X}_2 & \cdots & \tilde{X}_1\tilde{X}_m \\ \tilde{X}_2\tilde{X}_1 & \tilde{X}_2^2 & & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{X}_m\tilde{X}_1 & \tilde{X}_m\tilde{X}_2 & \cdots & \tilde{X}_m^2 \end{bmatrix}$$

$$= \begin{bmatrix} E(\tilde{X}_1^2) & E(\tilde{X}_1\tilde{X}_2) & \cdots & \\ E(\tilde{X}_2\tilde{X}_1) & & & \vdots \\ \vdots & & \ddots & \vdots \\ & \cdots & & E(\tilde{X}_m^2) \end{bmatrix}$$

$$= \begin{bmatrix} \text{Var}\,X_1 & \text{Cov}(X_1X_2) & \cdots & \cdots \\ \text{Cov}(X_1X_2) & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \vdots & \cdots & \cdots & \text{Var}\,X_m \end{bmatrix}$$

whereby,

$\text{Cov}(X_1, X_2) = E(X_1 - E(X_1))(X_2 - E(X_2))$

5. Matrix Equivalence Properties

vector	Matrix
a. $E(aX+b) = aE(X)+b$	a. $E(Ax+b) = AE(x)+b$
b. $Var(aX+b) = a^2 Var(X)$	b. $Var(Ax+b) = AVar(x)A^T$
c. $Var(X) = E(X^2) - [E(X)]^2$	c. $Var(x) = E(xx^T) - E(x)E(x)^T$

Proofs:

1. $E(Ax+b) = AE(x)+b$

- the ith element of the $k \times 1$ vector

$$(Ax+b)_i = \sum_{j=1}^{m} a_{ij}x_j + b_i$$

$$E(\Sigma a_{ij}X_j + b_i) = \sum_{j=1}^{m} a_{ij}E(X_j) + b_i$$

$$= (AE(x)+b)_i$$

2. $Var(Ax+b) = AVar(x)A^T$

- $Ax+b - E(Ax+b) = Ax+b - (AE(x)+b)$

$$= A[x - E(x)]$$

$$= A\tilde{x}$$

- $Var(Ax+b) = E[Ax+b - E(Ax+b)][Ax+b - E(Ax+b)]^T$

$$= E[A\tilde{x}][A\tilde{x}]^T$$

$$= AE(\tilde{x}\tilde{x}^T)A^T$$

$$= AVar(x)A^T$$

3. $Var(x) = E(xx^T) - E(x)E(x)^T$

$$Var(x) = E[(x - E(x)][(x - E(x)]^T$$

$$= E[(x - E(x))(x^T - E(x)^T)]$$

$$= E(xx^T - xE(x)^T - E(x)x^T + E(x)E(x)^T)$$

$$= E(xx^T) - E(x)E(x)^T - E(x)E(x^T) + E(x)E(x^T)$$

$$= E(xx^T) - E(x)E(x)^T$$

6. Linear Combination : $c^T x$

$$c = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}$$

$$c^T x = c_1 x_1 + c_2 x_2 + \dots + c_m x_m$$

$$= \sum c_i x_i$$

- ensuring variation, variables vary independently

$Var(c^T x) = c^T Var\, c \geq 0$ for all $c \neq 0_{m \times 1}$

- If $Var(c^T x) = 0$

a. One of the random variable is not random

b. one of the random variable is a multiple of the other

c. one of the random variable is a linear combination of two or more of the other random variables

Positive Definiteness

-if A is positive definite if

$c^T A c > 0$ for all $c \neq 0_{m \times 1}$

· positive semidefinite if $c^T A c \geq 0$ for all $c \neq 0_{m \times 1}$

- Application:

$Var(x)$ are positive semidefinite

If random variables in x are not linearly dependent, $Var(x) > 0$

Hence, $c^T Var(x) c > 0$

- Implication

If $c^T X^T X c = (Xc)^T Xc > 0$ for all $c \neq 0_{k \times 1}$ $Xc \neq 0_{n \times 1}$

Hence, $X^T X$ is positive definite

7. Sample Variance - Covariance Matrix

- allow X to be mean centered

- $\frac{1}{n-1} X^T X$ is the symmetric sample variance - covariance matrix

(5) Principal Component Analysis

$Ax = \lambda x$

- whereby, λ are eigenvalues

Eigendecomposition of symmetric matrix

- every kxk symmetric A can be decomposed

$$A = Q \Lambda Q^T$$

$$= [q_1 \dots q_k] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \dots & \dots & \lambda_k \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_k \end{bmatrix}$$

note $Q^T A Q = \Lambda$

- λ_i are eigenvalues (usually ranked $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$)
- $k \times 1$ q_i are eigenvectors
- Q satisfies the property $Q^T Q = I_k$

Application (PCA)

$$Var(x) = Q \Lambda Q^T$$

- allow us to check the definitness and variation in $Var(x)$
- Elaboration:

$$\frac{1}{n-1} X^T X = Q \Lambda Q^T$$

- if $Y = XQ$

Principal Component Analysis

$$\frac{1}{n-1} Y^T Y = \frac{1}{n-1} Q^T X^T X Q = Q^T \left(\frac{1}{n-1} X^T X\right) Q$$

$$= Q^T Q \Lambda Q^T Q$$

$$= \Lambda$$

- Q are principal components - representing linear combinations of the original variables in X. Each component captures a unique uncorrelated index variable or dimension which minimizes variance
- Components are ordered by eigenvalues

First Component of Q has the highest variance, meaning it captures the most variability in the data

Eg. $Q = \begin{bmatrix} q_{11} & q_{12} & q_{13} \\ q_{21} & q_{22} & q_{23} \\ q_{31} & q_{32} & q_{33} \end{bmatrix}$

$PC1 = q_{11} X_1 + q_{21} X_2 + q_{31} X_3$

see which has the highest q_{11}, q_{21} or q_{31}
represents importance of each X

(6) Matrix Algebra

Differentiation of Matrix

- let $f(x)$ be a function of $n \times 1$ vector x

$f(x) = x^T A x$ where A is an $n \times n$ matric

1. Gradient Vector [Scalar-valued Function]

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

$$\frac{\partial f}{\partial x^T} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_n} \end{bmatrix}$$

2. $y = Ax$ whereby A is not dependent on x

$$\frac{\partial y}{\partial x} = A$$

3. $y = Ax$ and $x = f(z)$

$$\frac{\partial y}{\partial z} = A \frac{\partial x}{\partial z}$$

4. $y = x^T A x$; $\frac{dy}{dx} = (A + A^T)x$ or $x^T(A + A^T)$

Proof: [Where $n = 2$]

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{pmatrix} a_{11}x_1 + a_{21}x_2 & a_{12}x_1 + a_{22}x_2 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= a_{11}x_1^2 + a_{21}x_1x_2 + a_{12}x_1x_2 + a_{22}x_2^2$$

$$\frac{\partial y}{\partial x_1} = 2a_{11}x_1 + a_{21}x_2 + a_{12}x_2$$

$$\frac{\partial y}{\partial x_2} = a_{21}x_1 + a_{12}x_1 + 2a_{22}x_2$$

$$\frac{dy}{dx} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} 2a_{11} & a_{12} + a_{21} \\ a_{21} + a_{12} & 2a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} 2a_{11}x_1 + a_{12}x_2 + a_{21}x_2 \\ a_{21}x_1 + a_{12}x_1 + 2a_{22}x_2 \end{bmatrix}$$

$$= [A + A^T]x$$

Additionally: If A is symmetric

$$\frac{dy}{dx} = (A + A^T)x$$

$$= 2Ax$$

5. $a = y^T x$ and $x = x(z)$ and $y = g(z)$

$$\frac{\partial a}{\partial z} = x^T \frac{\partial y}{\partial z} + y^T \frac{\partial x}{\partial z}$$

Jacobian Matrix

- matrix that captures all gradients of several functions wrt x
- allow x to be $n \times 1$ and $y = f_i(x) = \begin{bmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_m(x) \end{bmatrix}$

$$\text{Jacobian} = \frac{\partial y}{\partial x^T} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \frac{\partial f_1(x)}{\partial x_2} & \cdots & \frac{\partial f_1(x)}{\partial x_n} \\ \frac{\partial f_2(x)}{\partial x_1} & \frac{\partial f_2(x)}{\partial x_2} & \cdots & \vdots \\ \vdots & \vdots & & \vdots \\ \frac{\partial f_m(x)}{\partial x_1} & \frac{\partial f_m(x)}{\partial x_2} & \cdots & \frac{\partial f_m(x)}{\partial x_n} \end{bmatrix} \quad m \times n$$

Hessian Matrix

- matrix that compiles second order differentiation
- in this case $y = f(x)$

$$\frac{d}{dx^T}\left(\frac{dy}{dx}\right) = \frac{d}{dx^T}\begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \\ \vdots \\ \frac{\partial y}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 y}{\partial x_1^2} & \frac{\partial^2 y}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 y}{\partial x_n x_1} \\ \frac{\partial^2 y}{\partial x_1 \partial x_2} & \vdots & & \\ \vdots & \vdots & \ddots & \\ \frac{\partial^2 y}{\partial x_1 \partial x_n} & \cdots & \cdots & \frac{\partial^2 y}{\partial x_n^2} \end{bmatrix}$$

Optimization (min/max)

- stationary point : (x^*, y^*)

  whereby, $f'_x(x^*, y^*) = 0$

  $f'_y(x^*, y^*) = 0$

- Key Expressions

  a. Max Point [concave]

$$v_1^2 \frac{\partial^2 f(x,y)}{\partial x^2} + 2v_1 v_2 \frac{\partial^2 f(x,y)}{\partial x \partial y} + v_2^2 \frac{\partial^2 f(x,y)}{\partial y^2} < 0 \quad \text{for all } v_1, v_2 \neq 0$$

  b. Min Point [convex]

$$v_1^2 \frac{\partial^2 f(x,y)}{\partial x^2} + 2v_1 v_2 \frac{\partial^2 f(x,y)}{\partial x \partial y} + v_2^2 \frac{\partial^2 f(x,y)}{\partial y^2} > 0 \quad \text{for all } v_1, v_2 \neq 0$$

- Quadratic Form Expressions

$$[v_1 \;\; v_2] \begin{bmatrix} \frac{\partial^2 f(y,x)}{\partial x^2} & \frac{\partial^2 f(y,x)}{\partial x \partial y} \\ \frac{\partial^2 f(y,x)}{\partial x \partial y} & \frac{\partial^2 f(y,x)}{\partial y^2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

  - Negative definiteness $v^T H_f v < 0$ [concave — Max Point]
  - Positive definiteness $v^T H_f v > 0$ [convex — Min Point]
  - At the stationary point:

$$\frac{\partial f(x^*, y^*)}{\partial x} = 0$$

$$\frac{\partial f(x^*, y^*)}{\partial y} = 0$$